GHAZI UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE

PROGRAMMING FUNDAMENTALS

LAB #03

LABORATORY ASSIGNMENT

STUDENT NAME

Saba Ashraf

ROLL NUMBER

24

PROGRAM

BS Computer Science

SECTION

A

SUBMITTED TO

Sir Asad Majeed

Day:

Lab - 3

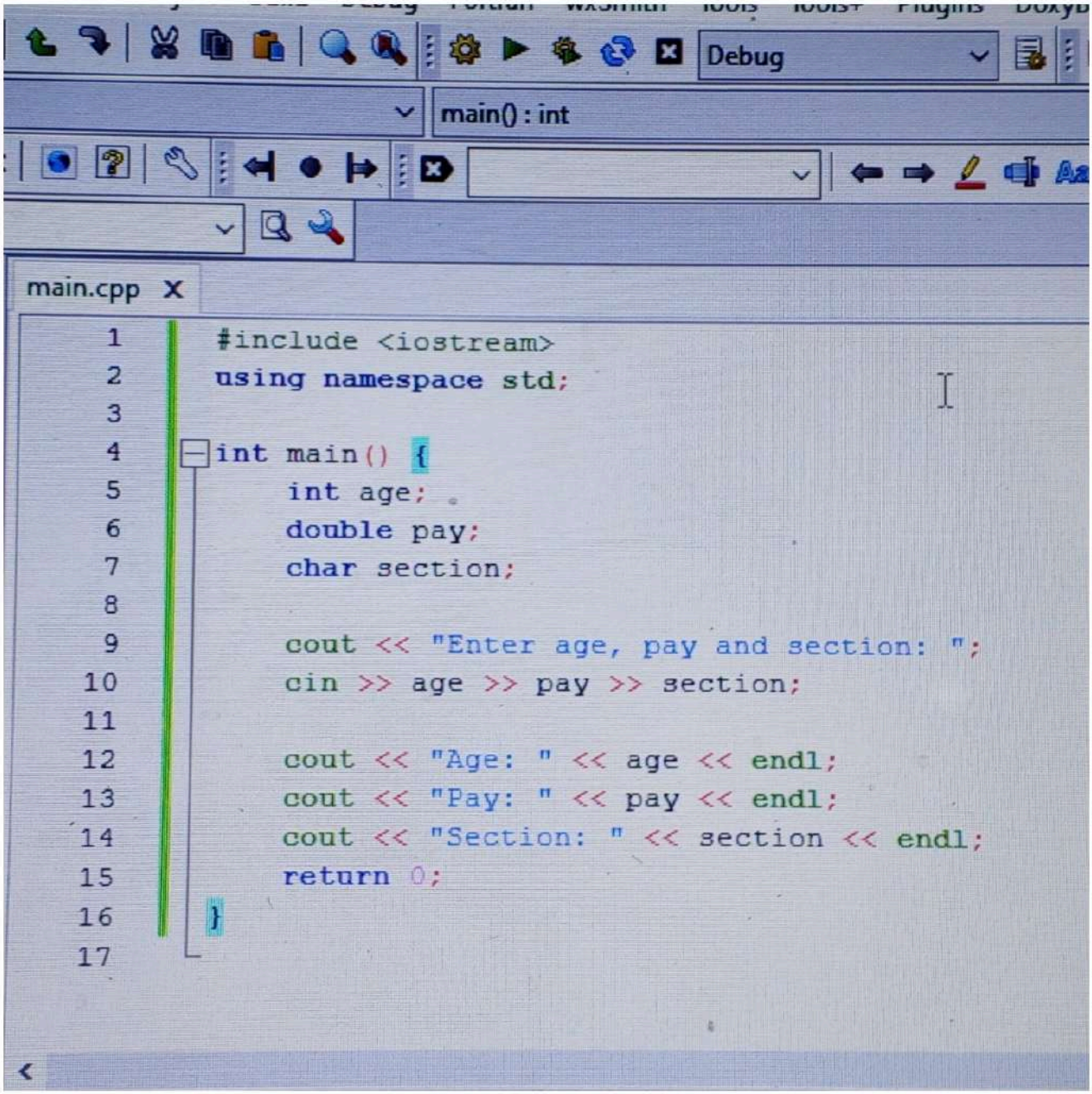
Date:

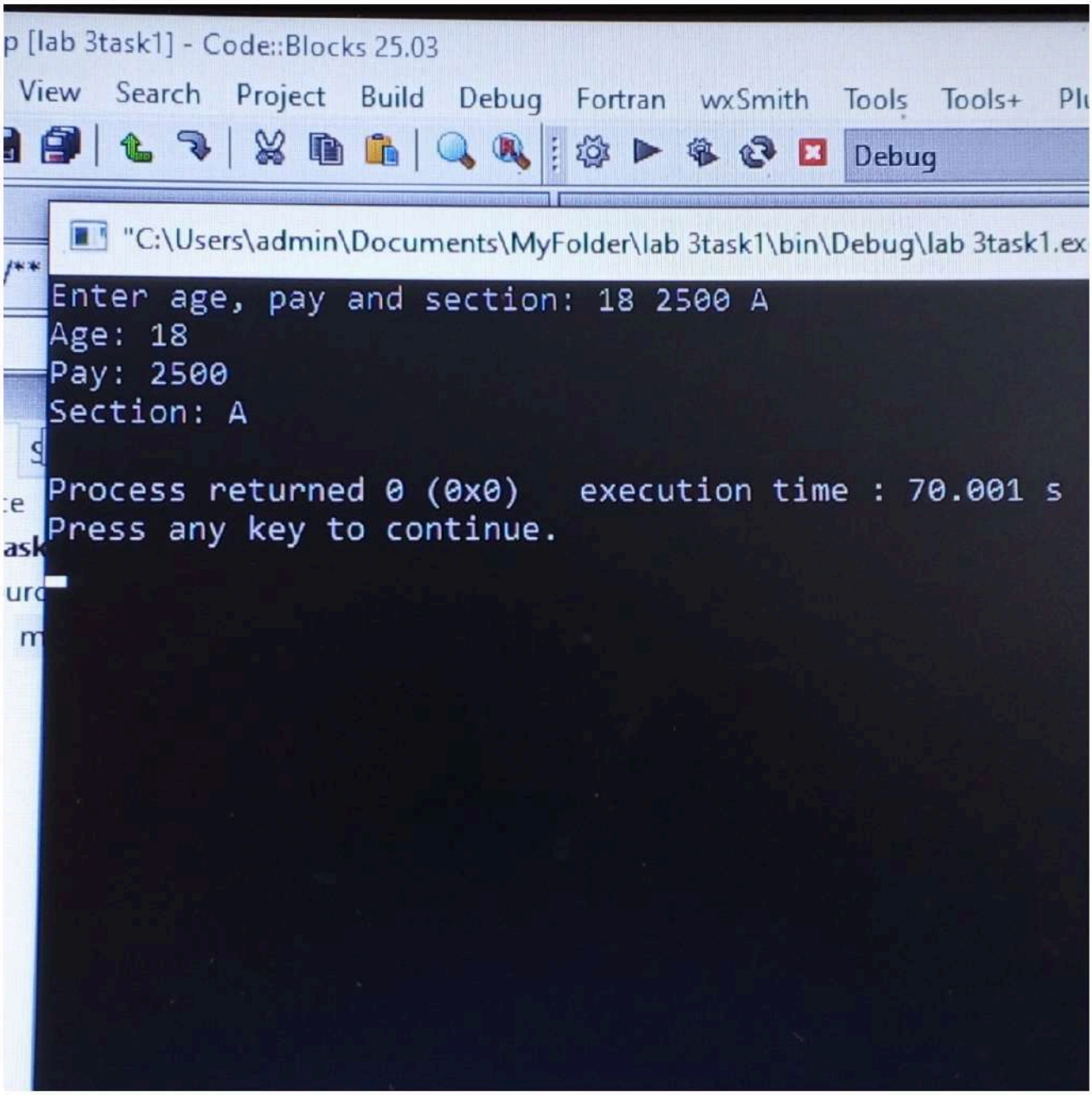
Task - 1

```
#include <iostream>
using namespace std;
int main ( ) {
    int age ;
    double pay ;
    char section;
   cout << " Enter age, pay and section: ";
    cin >> age >> pay >> section;
    cout << "Age :" << age << endl ;
    cout << "pay:" << pay << endl;
    cout << "section:" << section << endl;
    return 0 ;
}
```

Output :

```
Enter age, pay and section : 18 2500 A
Age : 18
Pay : 2500
section: A
```


Debug
main() : int
main.cpp
#include <iostream>
using namespace std;
int main() {
int age;
double pay;
char section;
cout << "Enter age, pay and section: ";
cin >> age >> pay >> section;
cout << "Age: " << age << endl;
cout << "Pay: " << pay << endl;
cout << "Section: " << section << endl;
return 0;
}

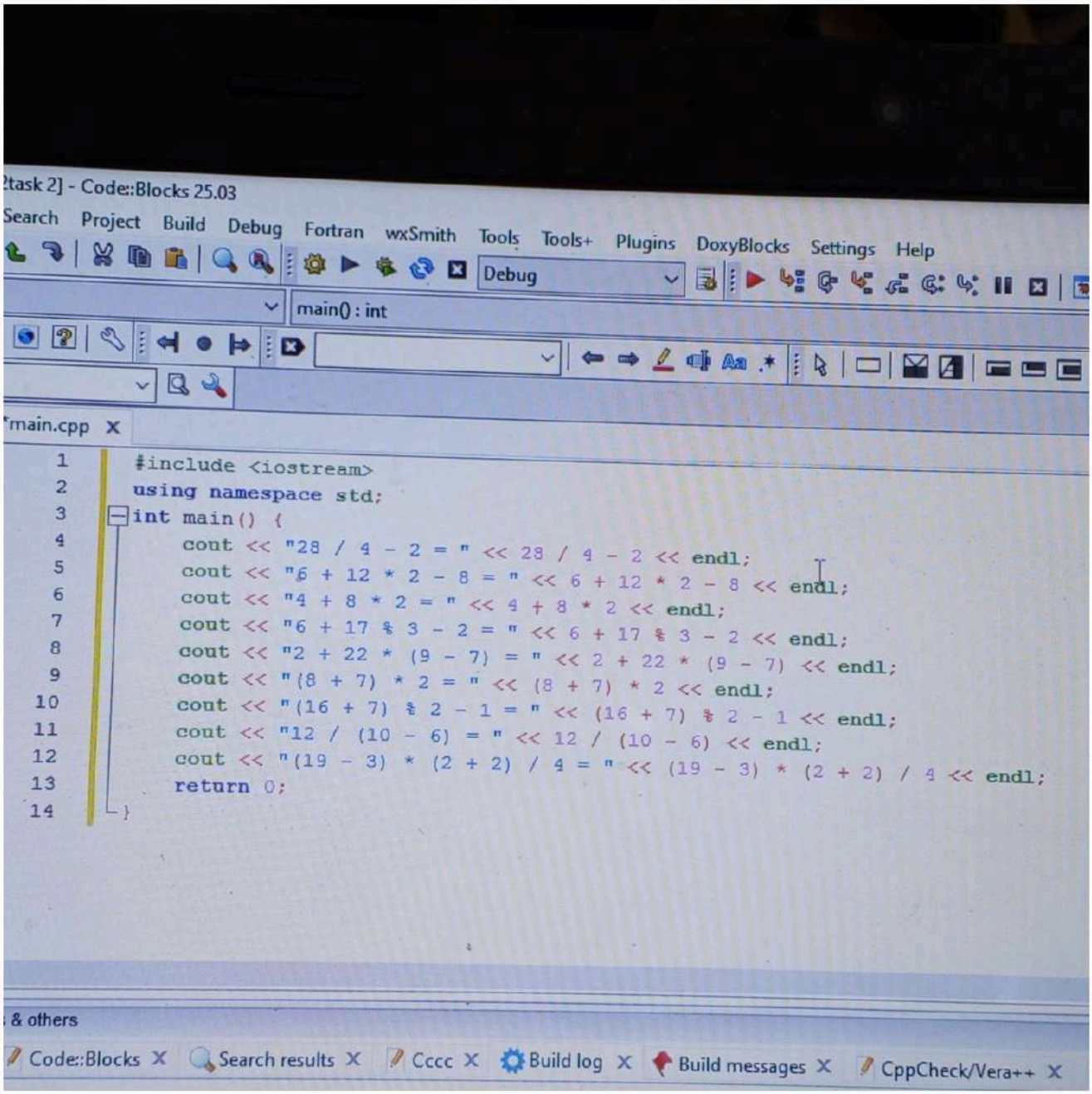

p [lab 3task1] - Code::Blocks 25.03
View Search Project Build Debug Fortran wxSmith Tools Tools+
Debug
"C:\Users\admin\Documents\MyFolder\lab 3task1\bin\Debug\lab 3task1.ex
Enter age, pay and section: 18 2500 A
Age: 18
Pay: 2500
Section: A
Process returned 0 (0x0) execution time : 70.001 s
Press any key to continue.

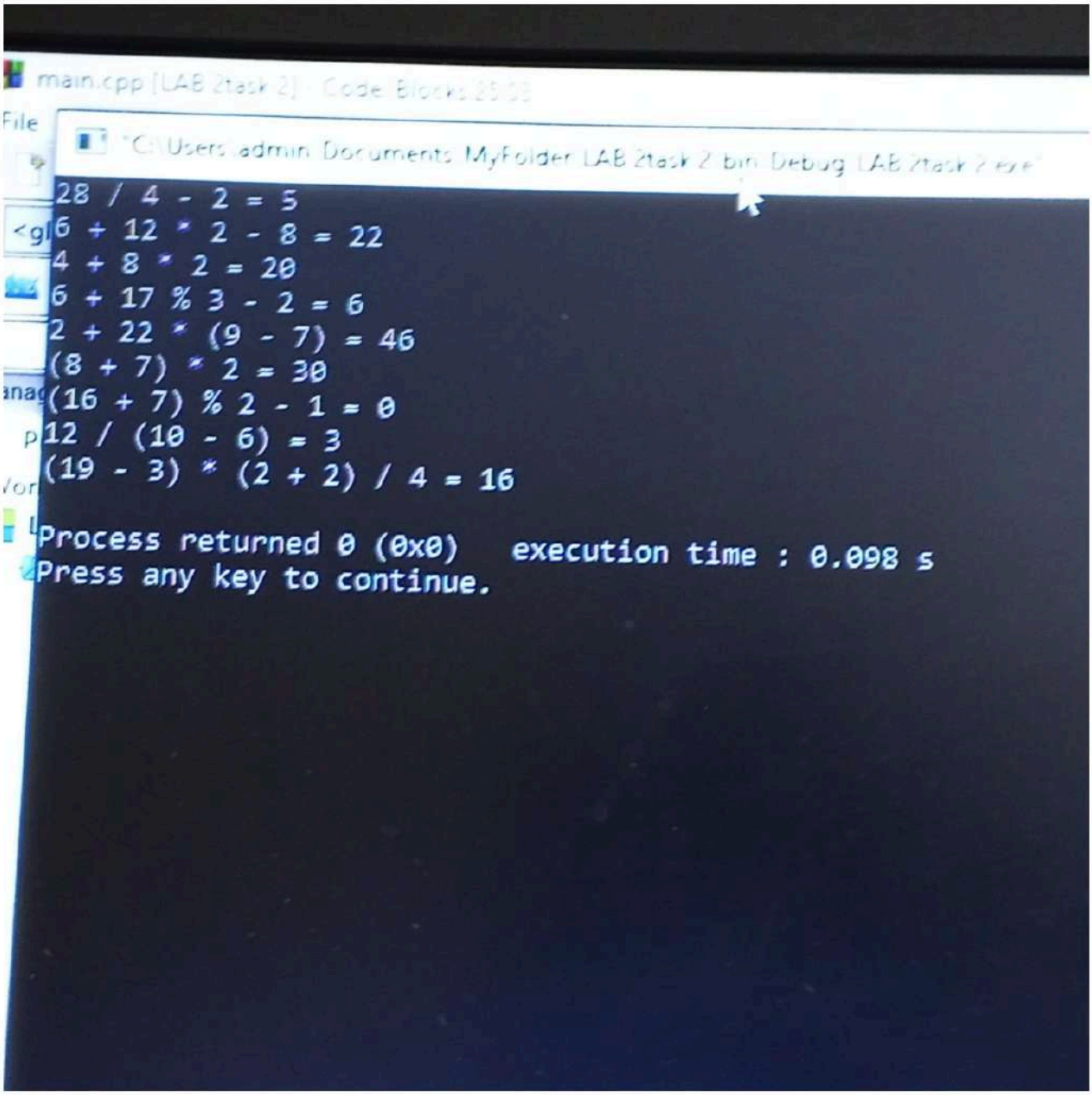
Lab - 03 Task - 02

```
#include <iostream>
using namespace std;
int main ()
{
    cout<<" 28/4 - 2 =" <<28/4-2<< endl;
    cout<<"6 + 12*2-8=" <<6 +12 *2-8<<endl;
    cout<<"4+8*2 = " << 4+8* 2 <<end l;
    cout <<"6+17%3-2=" <<6+17%3-2 <<endl;
    cout<<" 2+22*(9-7) = "<< 2+22*(9-7)<<endl;
    cout<<"(8+7)*2 = "<< (8+7) * 2 <<endl;
    cout<<"(16+7)%2-1 = "<< (16+7)%2-1<<endl;
    cout<<"12/(10-6) = " << 12/(10-6) <<endl;
    cout <<"(19-3)*(2+2)/4 ="<<(19-3)*(2+2)/4<<endl;
    return 0;
}
```

Output

```
28/4 - 2 = 5
6+12*2 - 8 = 22
4 +8 *2  = 20
6 + 17%3 - 2 = 6
2 + 22 *(9-7)= 46
(8+7) * 2  = 30
(16+7) %2  -1 = 0
12 /(10-6) =  3
(19-3) * ( 2+2) /4 = 16
```


task 2] - Code::Blocks 25.03
Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help
Debug
main() : int
*main.cpp
#include <iostream>
using namespace std;
int main() {
cout << "28 / 4 - 2 = " << 28 / 4 - 2 << endl;
cout << "6 + 12 * 2 - 8 = " << 6 + 12 * 2 - 8 << endl;
cout << "4 + 8 * 2 = " << 4 + 8 * 2 << endl;
cout << "6 + 17 % 3 - 2 = " << 6 + 17 % 3 - 2 << endl;
cout << "2 + 22 * (9 - 7) = " << 2 + 22 * (9 - 7) << endl;
cout << "(8 + 7) * 2 = " << (8 + 7) * 2 << endl;
cout << "(16 + 7) % 2 - 1 = " << (16 + 7) % 2 - 1 << endl;
cout << "12 / (10 - 6) = " << 12 / (10 - 6) << endl;
cout << "(19 - 3) * (2 + 2) / 4 = " << (19 - 3) * (2 + 2) / 4 << endl;
return 0;
}
& others
Code::Blocks
Search results
Cccc
Build log
Build messages
CppCheck/Vera++


main.cpp [LAB 2task 2] - Code Blocks 25.03
File
"C:\Users\admin\Documents\MyFolder\LAB 2task 2\bin\Debug\LAB 2task 2.exe"
28 / 4 - 2 = 5
6 + 12 * 2 - 8 = 22
4 + 8 * 2 = 20
6 + 17 % 3 - 2 = 6
2 + 22 * (9 - 7) = 46
(8 + 7) * 2 = 30
(16 + 7) % 2 - 1 = 0
12 / (10 - 6) = 3
(19 - 3) * (2 + 2) / 4 = 16
Process returned 0 (0x0) execution time : 0.098 s
Press any key to continue.

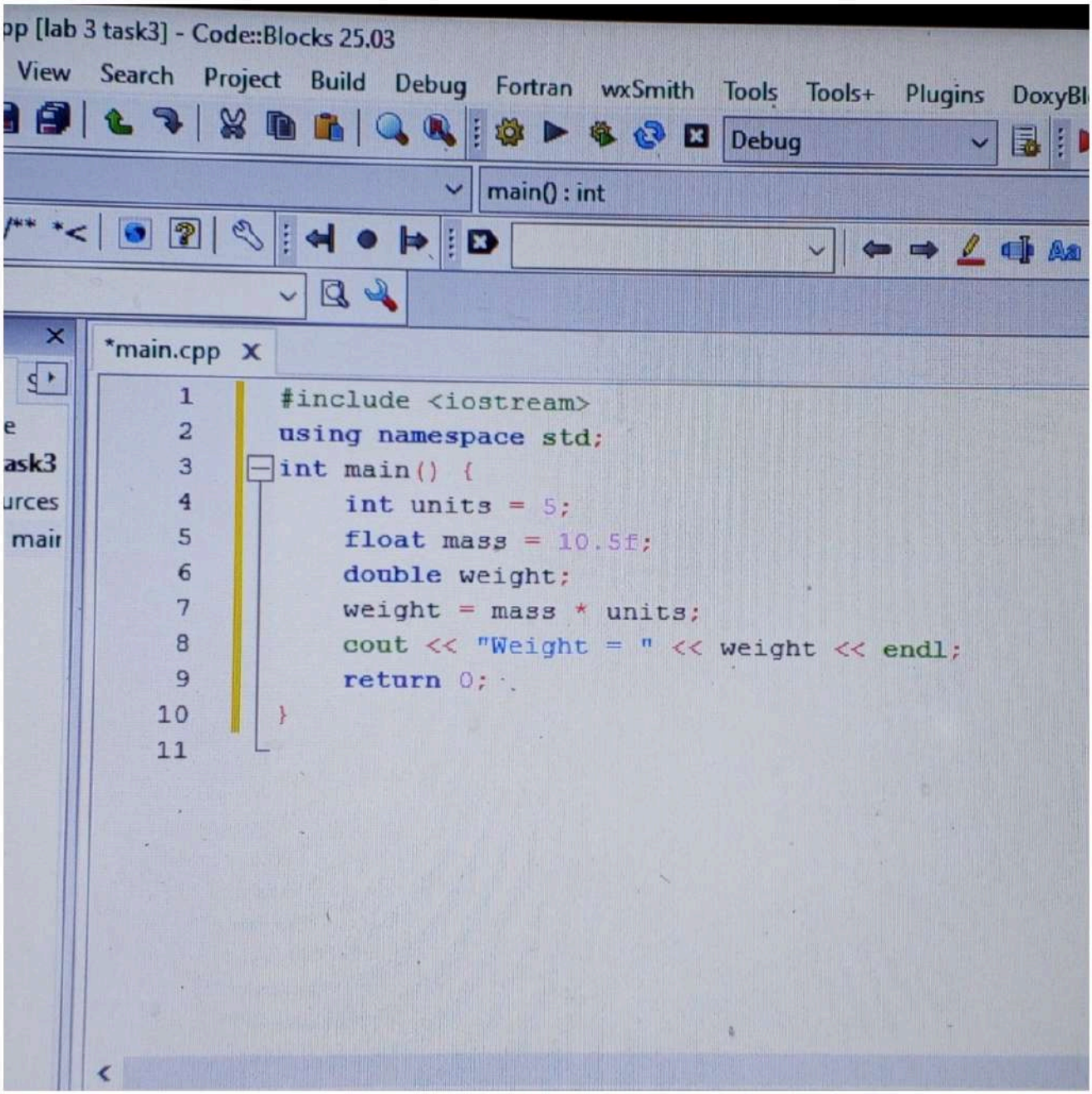
Date:

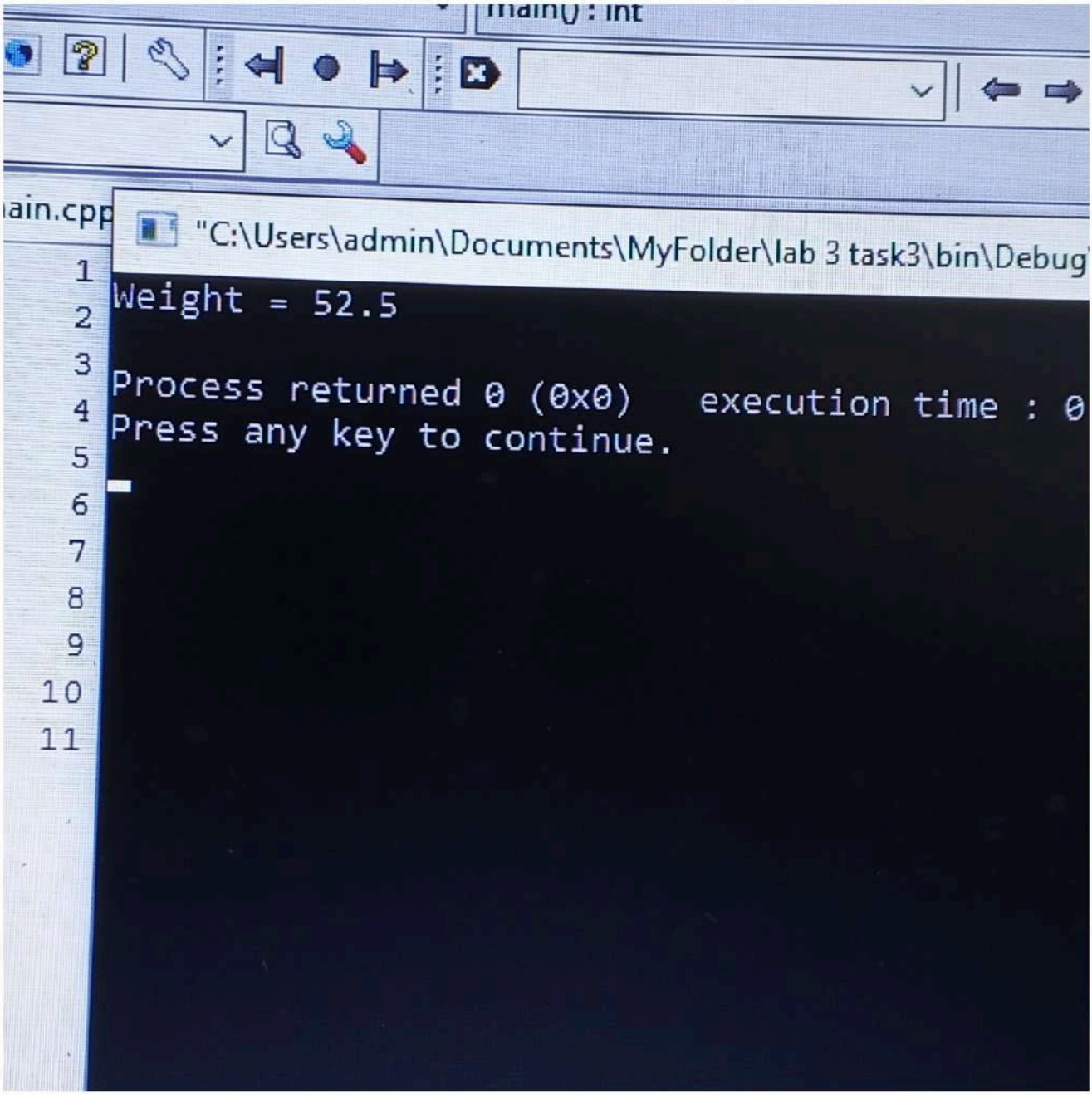
Lab-03 Task - 03

```
# include <iostream>
using namespace std;
int main() {
        int units = 5;
        Float mass = 10.5f;
        double weight ;
        weight = mass * units
        Cout << "weight = " << weight <<endl
    return 0;
}
```

output

weight = 52.5


pp [lab 3 task3] - Code::Blocks 25.03
View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins
Debug
main() : int
*main.cpp
1 #include <iostream>
2 using namespace std;
3 int main() {
4 int units = 5;
5 float mass = 10.5f;
6 double weight;
7 weight = mass * units;
8 cout << "Weight = " << weight << endl;
9 return 0;
10 }
11

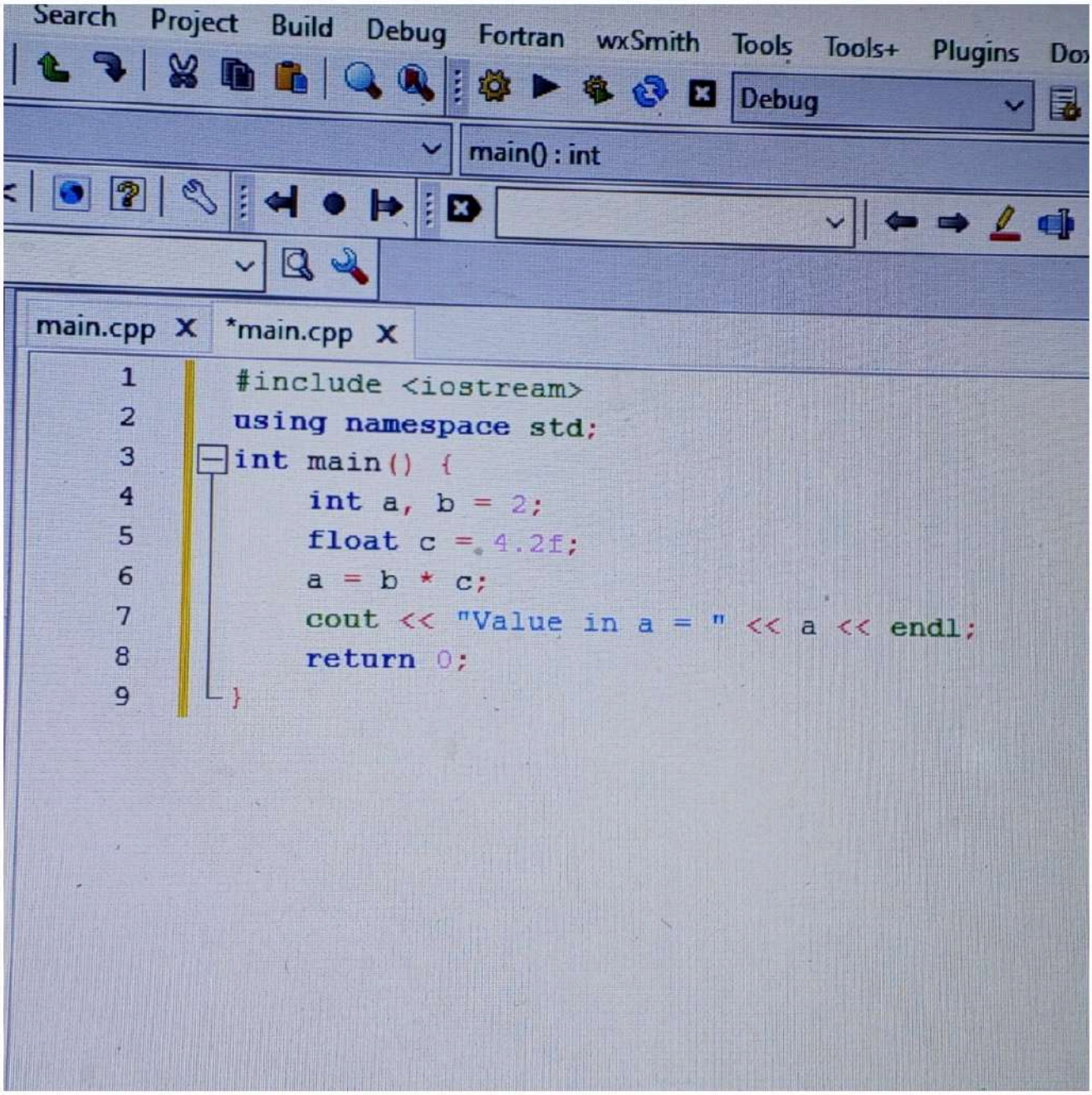

main() : int
ain.cpp
"C:\Users\admin\Documents\MyFolder\lab 3 task3\bin\Debug
1
2
3
4
5
6
7
8
9
10
11
Weight = 52.5
Process returned 0 (0x0)   execution time : 0
Press any key to continue.

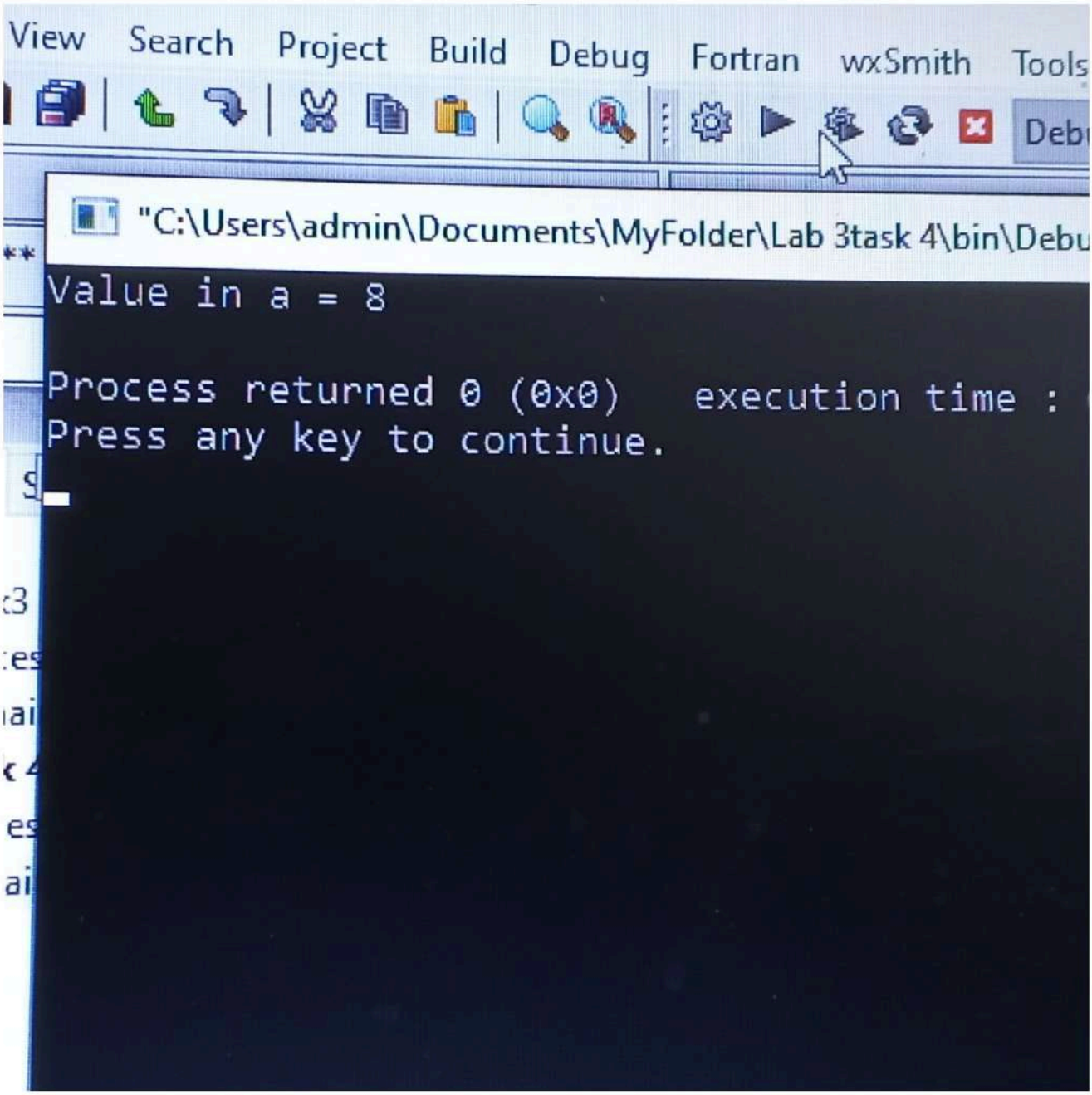
Lab-03 Task- 04

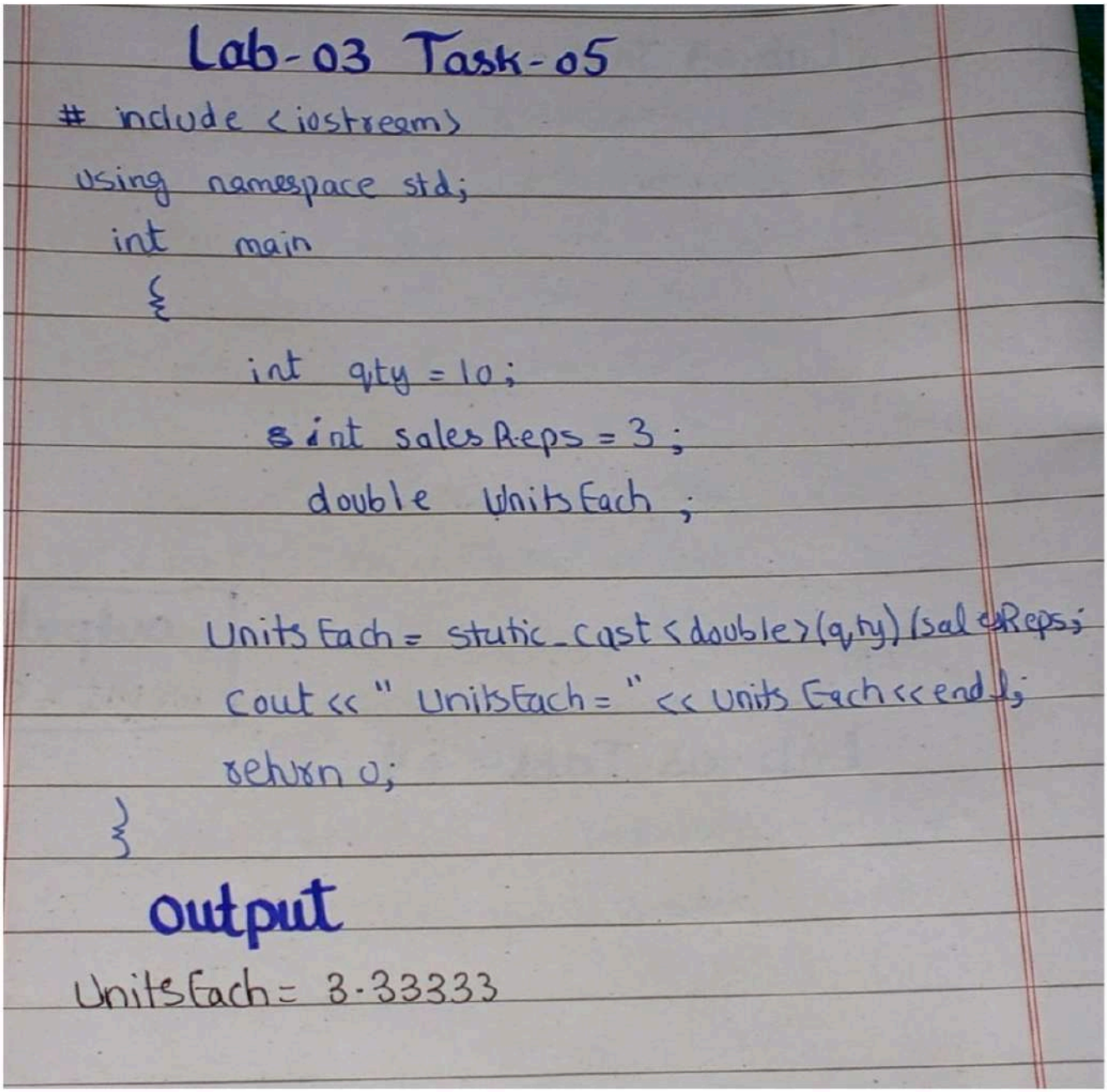
```
#include<iostream>
using namespace std;
int main() {
    int a, b = 2;
    float c = 4.2f;
    a = b * c;
cout << "value in a = " << a << endl;
    return 0;
}
```

Output

```
Value in a = 8
```


Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins
Debug
main() : int
main.cpp
*main.cpp
#include <iostream>
using namespace std;
int main() {
int a, b = 2;
float c = 4.2f;
a = b * c;
cout << "Value in a = " << a << endl;
return 0;
}


View Search Project Build Debug Fortran wxSmith Tools
"C:\Users\admin\Documents\MyFolder\Lab 3task 4\bin\Debu
Value in a = 8
Process returned 0 (0x0) execution time :
Press any key to continue.



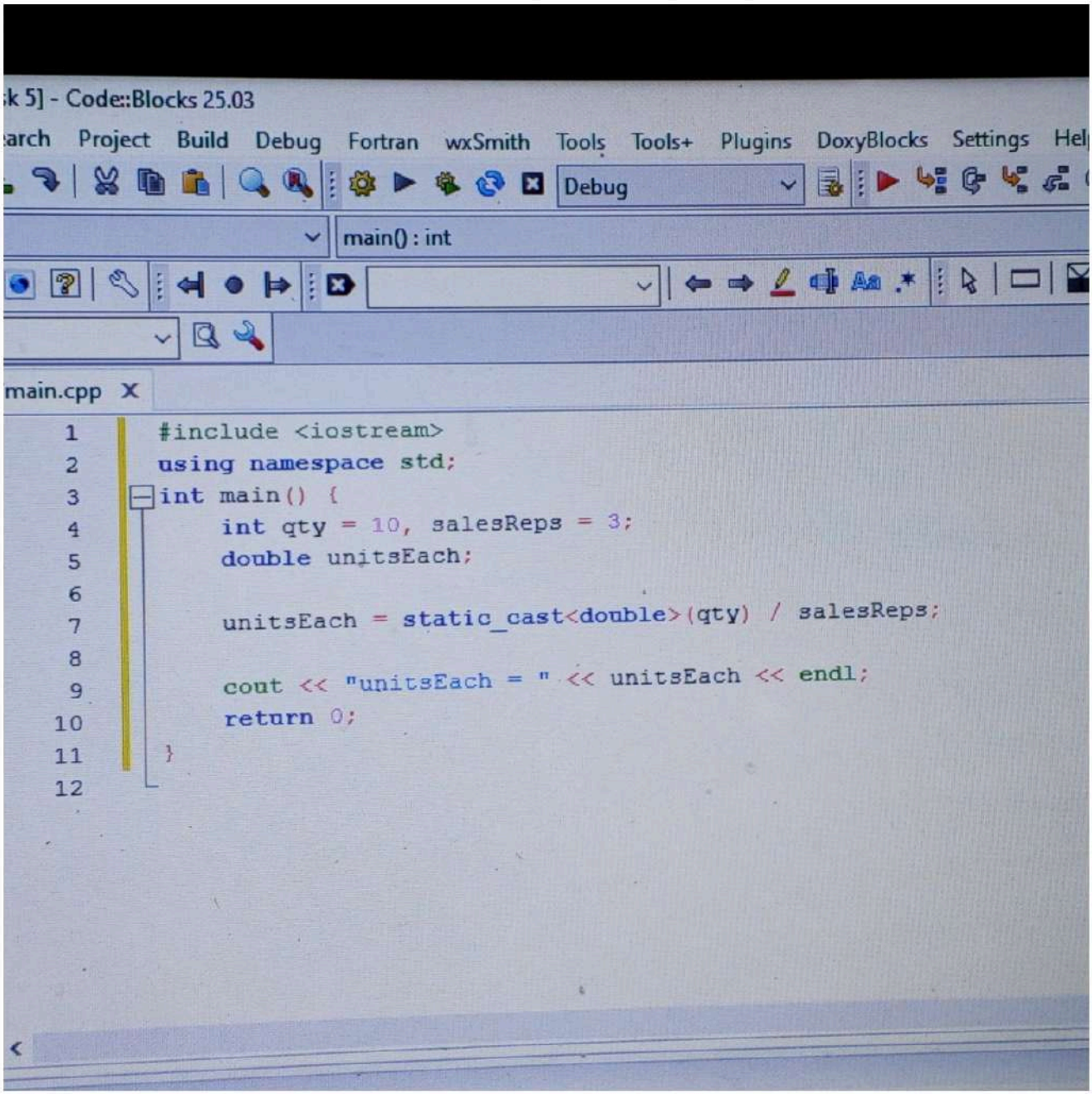
Lab-03 Task-05

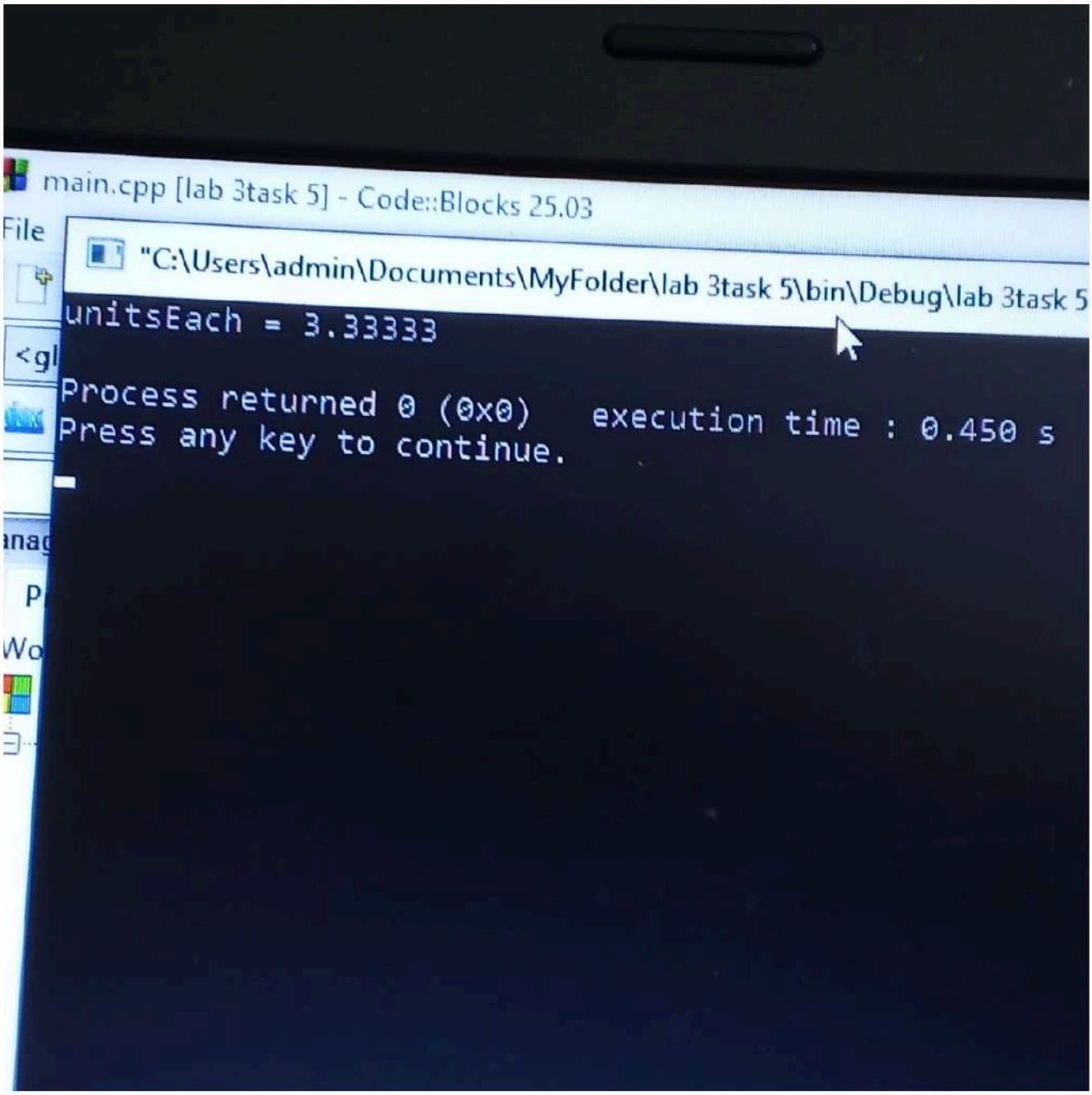
```
# include <iostream>
using namespace std;
 int  main
  {
      int qty = 10;
      int salesReps = 3;
       double UnitsEach ;

    UnitsEach = stutic_cast<double>(qty)/salesReps;
    Cout << " UnitsEach = " << UnitsEach << endl;
    return 0;
}
```

Output

UnitsEach = 3.33333


k 5] - Code::Blocks 25.03
arch Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Hel
Debug
main() : int
main.cpp X
1 #include <iostream>
2 using namespace std;
3 int main() {
4 int qty = 10, salesReps = 3;
5 double unitsEach;
6
7 unitsEach = static_cast<double>(qty) / salesReps;
8
9 cout << "unitsEach = " << unitsEach << endl;
10 return 0;
11 }
12


main.cpp [lab 3task 5] - Code::Blocks 25.03
File
"C:\Users\admin\Documents\MyFolder\lab 3task 5\bin\Debug\lab 3task 5
unitsEach = 3.33333
Process returned 0 (0x0)   execution time : 0.450 s
Press any key to continue.

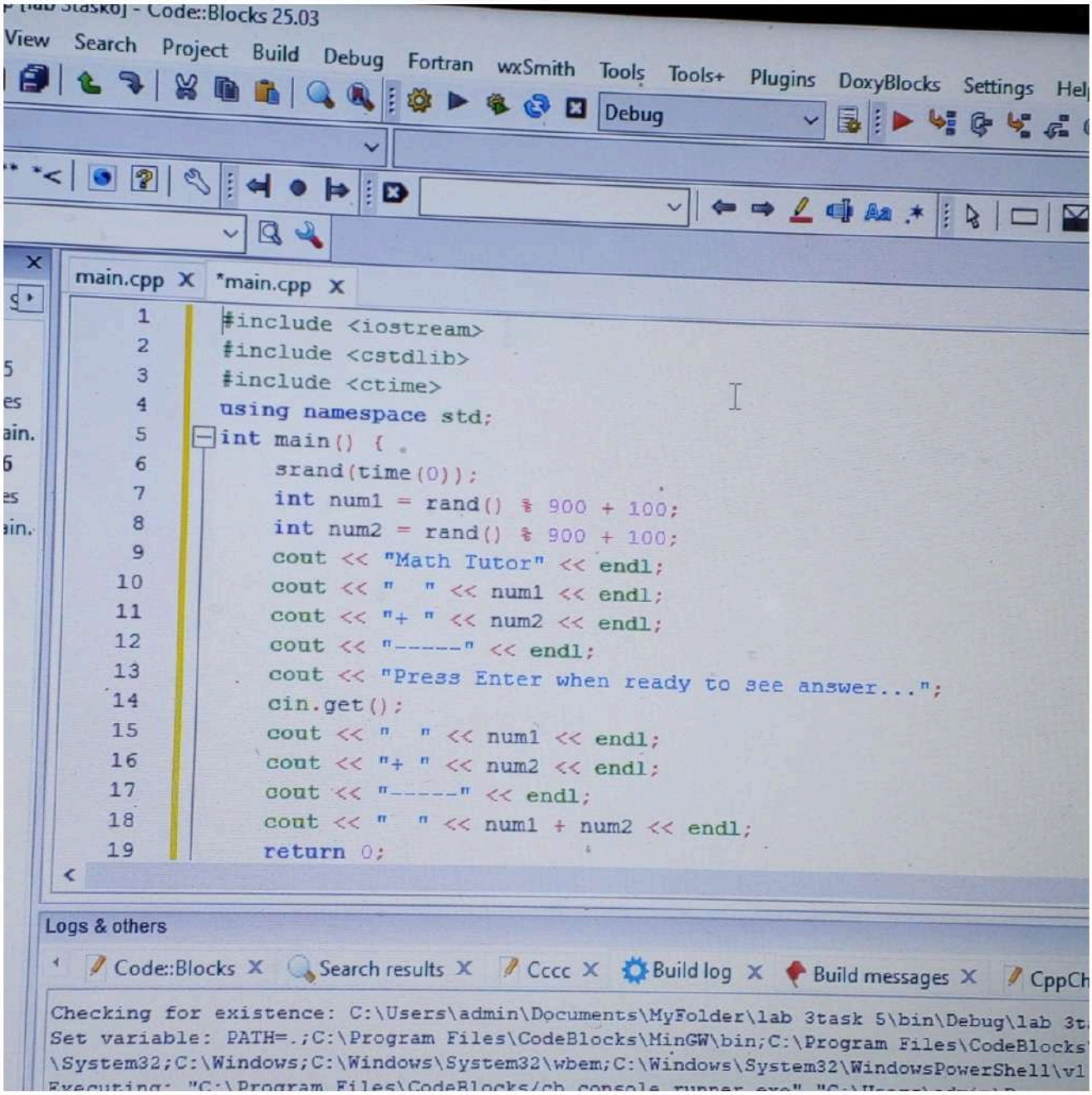
Date:

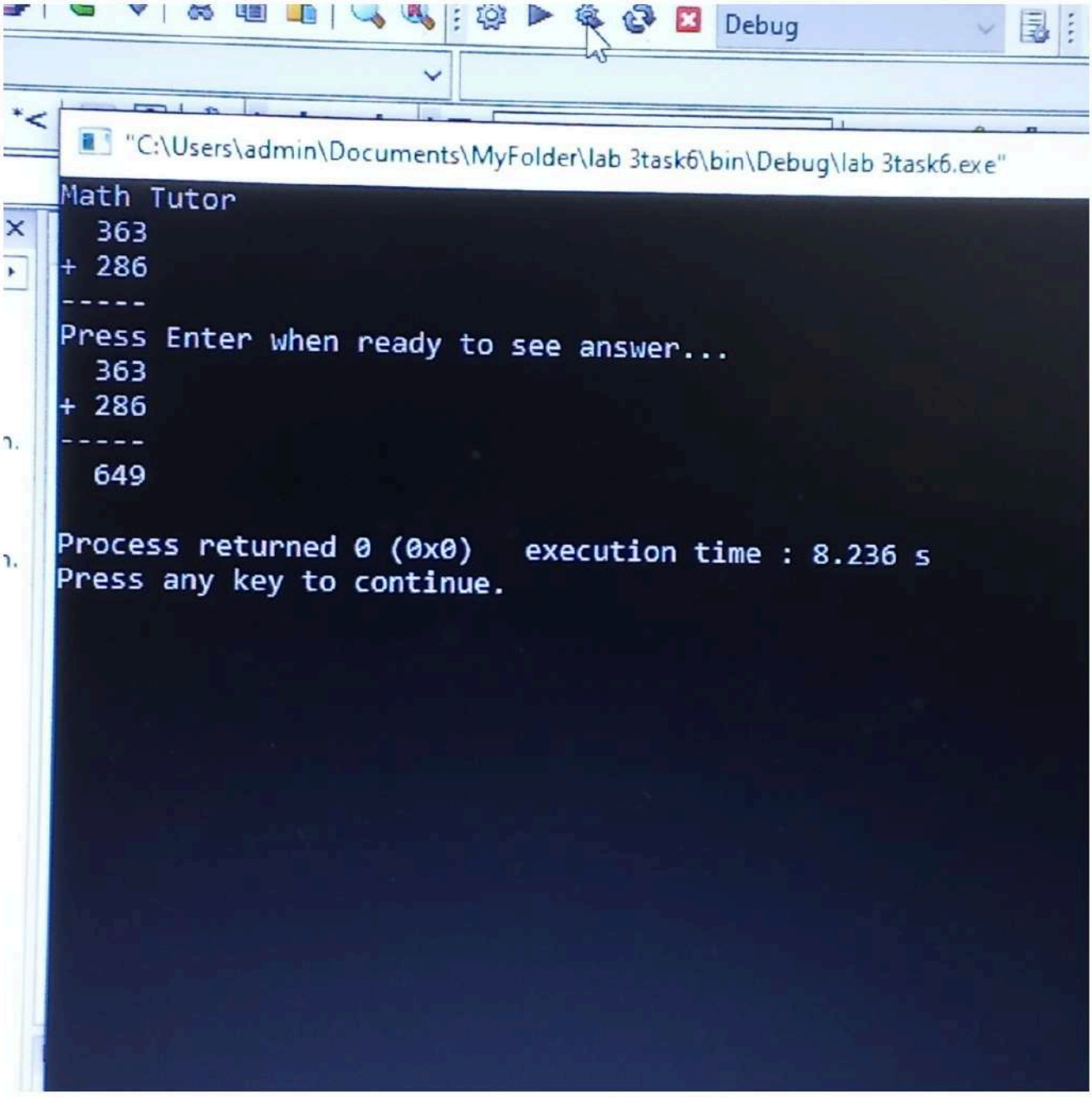
Lab-03 Task-06

```
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;
int main()
{
    srand(time(0));
    int num1 = rand()%900+100;
    int num2 = rand()%900+100;
  cout<<"Math Tutor"<<endl;
   cout<<"  "<<num1<<endl;
   cout<<" + "<<num)<<endl;
   cout<<"_____"<<endl;
  cout<<"Press Enter when ready to see answer...";
    cin.get();
   cout<<"  "<<num1<<endl;
   cout<<" + "<<num2<<endl;
   cout<<"_____"<<endl;
   cout<<"  "<<num1+num2<<endl;
   return 0;
}
```

Output:

```
Math Tutor
 363
+ 286
.....
 363
+_286
 ---
 649
```


Code::Blocks 25.03
View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings
Debug
main.cpp
*main.cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;
int main() {
srand(time(0));
int num1 = rand() % 900 + 100;
int num2 = rand() % 900 + 100;
cout << "Math Tutor" << endl;
cout << " " << num1 << endl;
cout << "+ " << num2 << endl;
cout << "-----" << endl;
cout << "Press Enter when ready to see answer...";
cin.get();
cout << " " << num1 << endl;
cout << "+ " << num2 << endl;
cout << "-----" << endl;
cout << " " << num1 + num2 << endl;
return 0;
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messages
Checking for existence: C:\Users\admin\Documents\MyFolder\lab 3task 5\bin\Debug\lab 3t
Set variable: PATH=.;C:\Program Files\CodeBlocks\MinGW\bin;C:\Program Files\CodeBlocks
\System32;C:\Windows;C:\Windows\System32\wbem;C:\Windows\System32\WindowsPowerShell\v1


Debug
"C:\Users\admin\Documents\MyFolder\lab 3task6\bin\Debug\lab 3task6.exe"
Math Tutor
363
+ 286

Press Enter when ready to see answer...
363
+ 286

649
Process returned 0 (0x0) execution time : 8.236 s
Press any key to continue.

ay: Date:

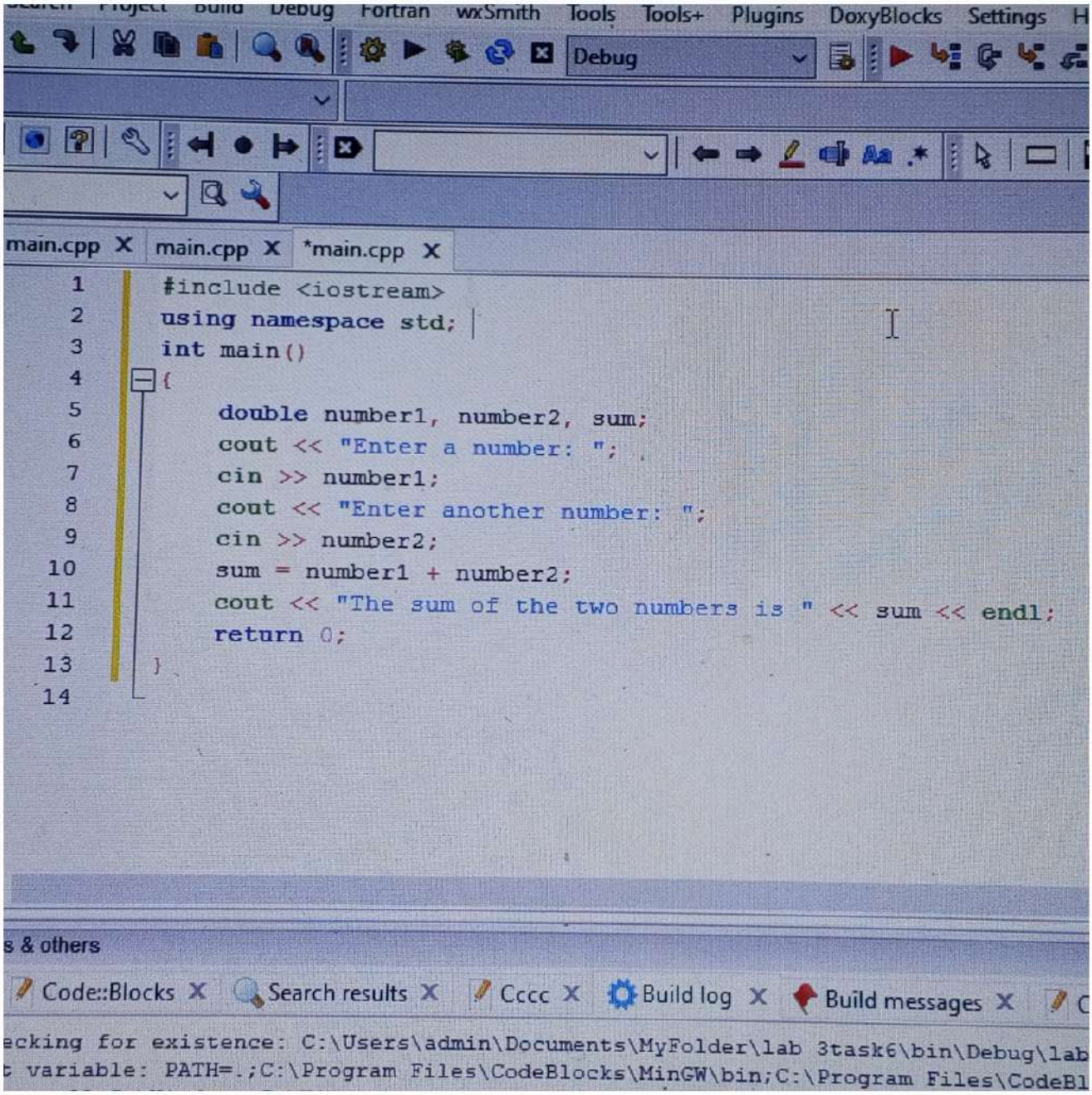
Lab-03 Home Task 1

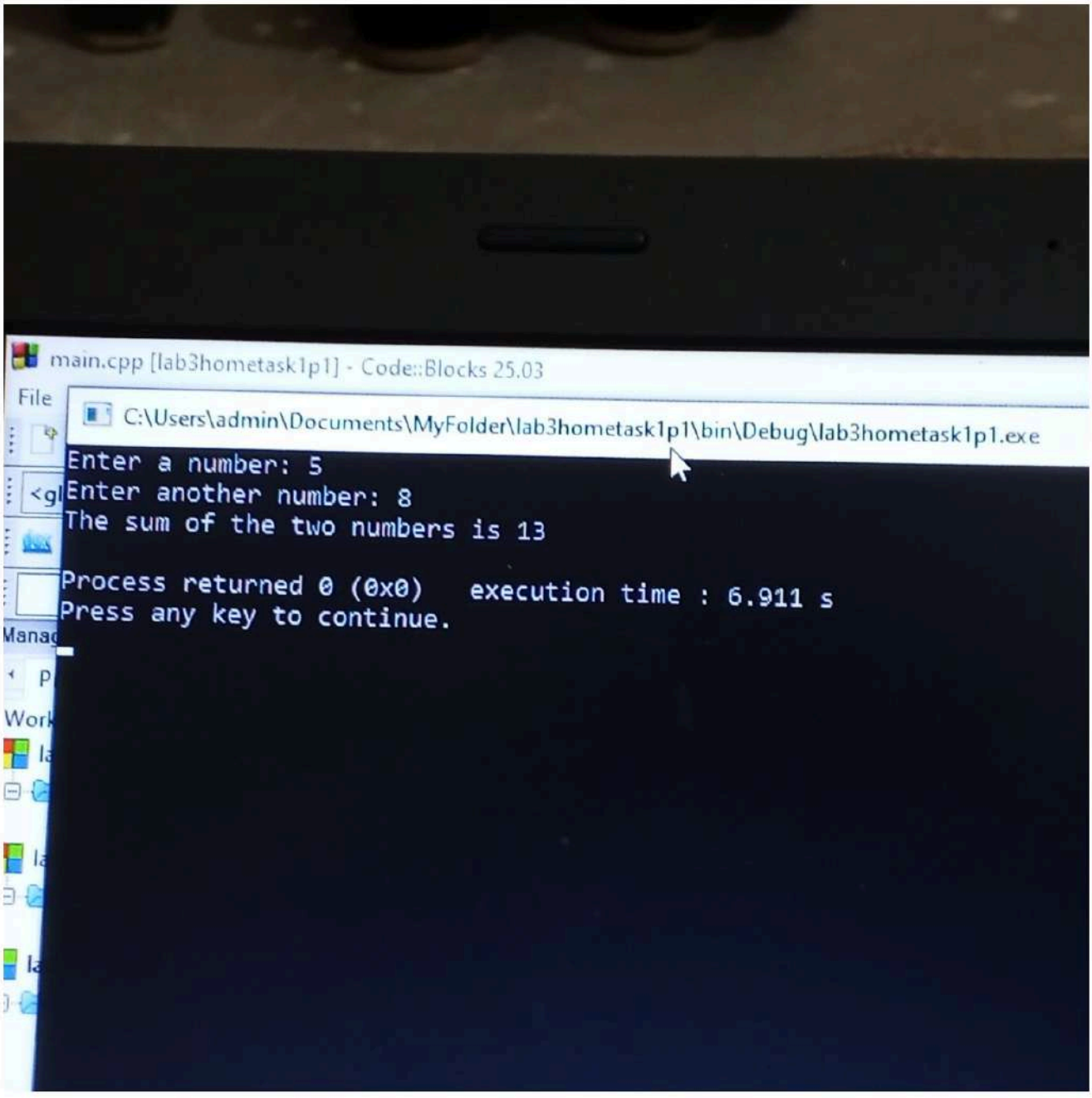
Program 1:

```
#include<iostream>
using namespace std;
int main() {
    double number1, number2, sum;
    cout<<"Enter a number: ";
    cin >> number1;
    cout<<" Enter another number: ";
    cin >> number2;
    sum = number1 + number2;
cout<<"The sum of the two numbers is "<<sum<<endl;
return 0;
}
```

Output

```
Enter a number: 5
Enter another number: 8
The sum of the two number is 13
```


Debug
main.cpp
main.cpp
*main.cpp
#include <iostream>
using namespace std;
int main()
{
double number1, number2, sum;
cout << "Enter a number: ";
cin >> number1;
cout << "Enter another number: ";
cin >> number2;
sum = number1 + number2;
cout << "The sum of the two numbers is " << sum << endl;
return 0;
}
Code::Blocks
Search results
Cccc
Build log
Build messages


main.cpp [lab3hometask1p1] - Code::Blocks 25.03
File
C:\Users\admin\Documents\MyFolder\lab3hometask1p1\bin\Debug\lab3hometask1p1.exe
Enter a number: 5
Enter another number: 8
The sum of the two numbers is 13
Process returned 0 (0x0) execution time : 6.911 s
Press any key to continue.

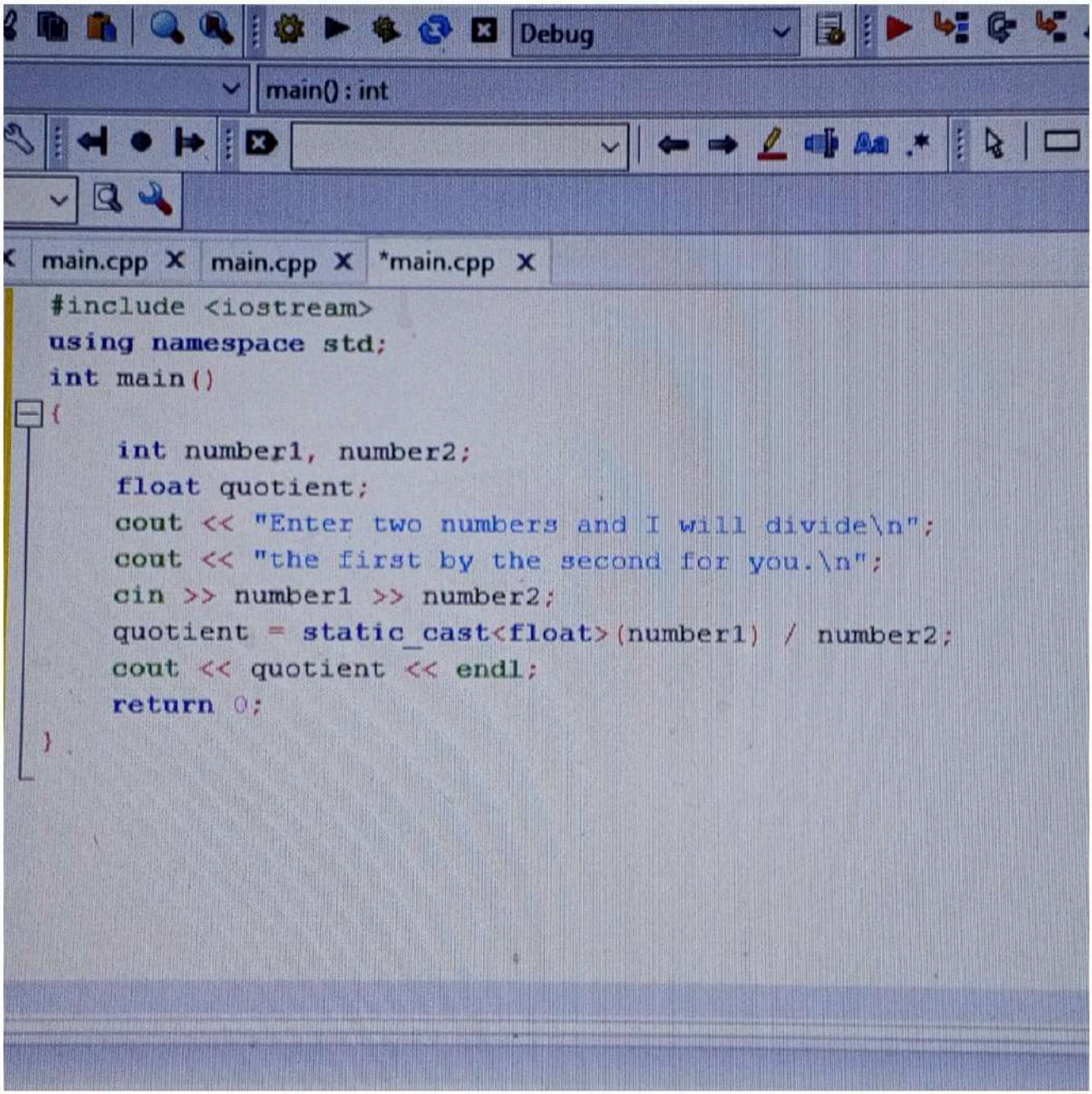
Home Task- 01

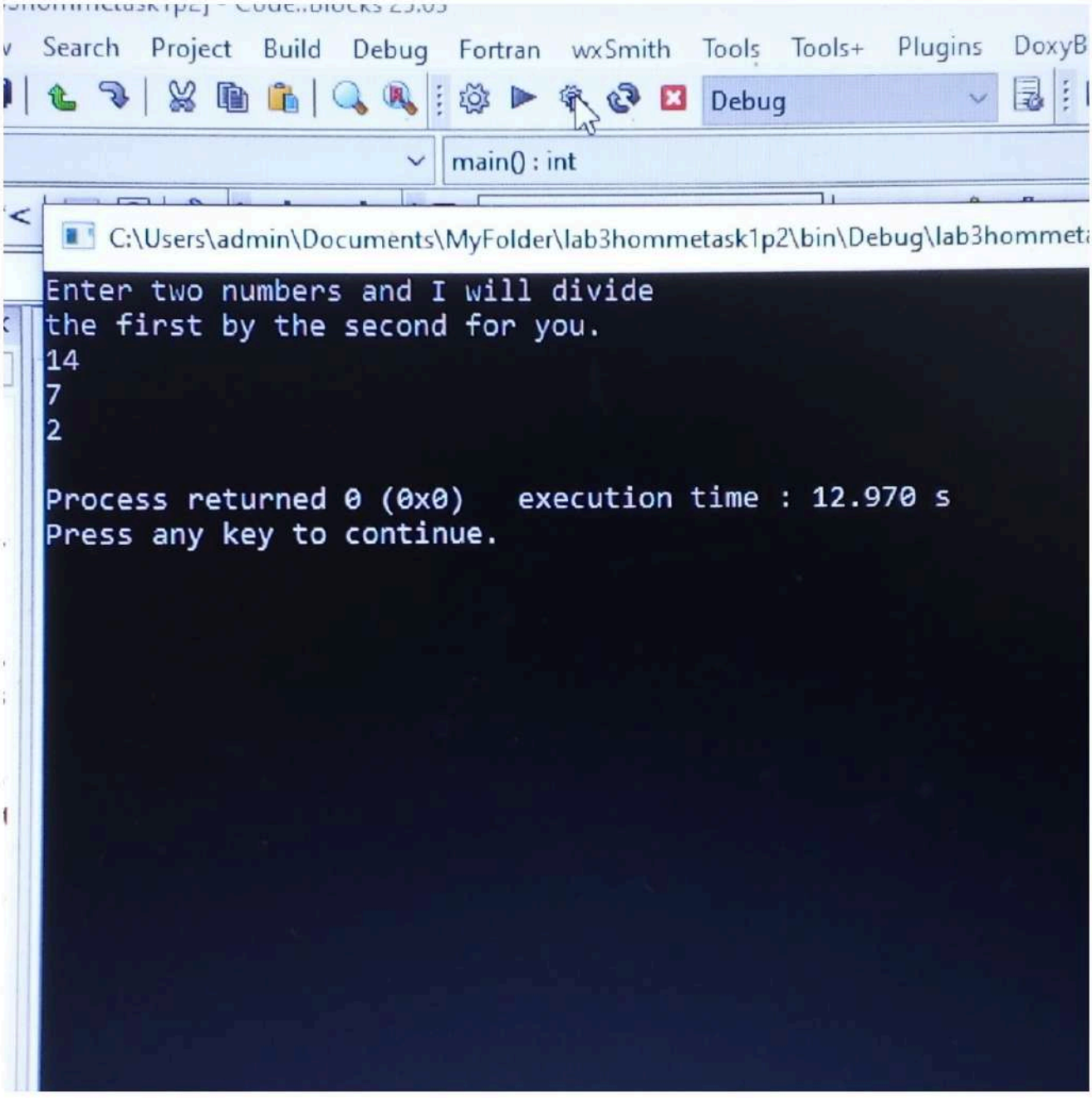
Program 2:-

```
#include <iostream>
Using namespace std;
int main()
{
    int number 1, number 2;
     float quotient ;
    cout << " Enter two numbers and I will divide\n";
   cout << " the first by the second for you.\n";
    cin >> number 1 >> number 2;
     quotient = static_cast<float>(number1)/ number 2;
     cout << quotient << endl;
     return 0;
}
```

Output

```
Enter two numbers and I will divide
the first by the second for you
14
7
2
```


Debug
main() : int
main.cpp
main.cpp
*main.cpp
#include <iostream>
using namespace std;
int main()
{
int number1, number2;
float quotient;
cout << "Enter two numbers and I will divide\n";
cout << "the first by the second for you.\n";
cin >> number1 >> number2;
quotient = static_cast<float>(number1) / number2;
cout << quotient << endl;
return 0;
}


Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyB
Debug
main() : int
C:\Users\admin\Documents\MyFolder\lab3hommetask1p2\bin\Debug\lab3hommeta
Enter two numbers and I will divide
the first by the second for you.
14
7
2
Process returned 0 (0x0) execution time : 12.970 s
Press any key to continue.

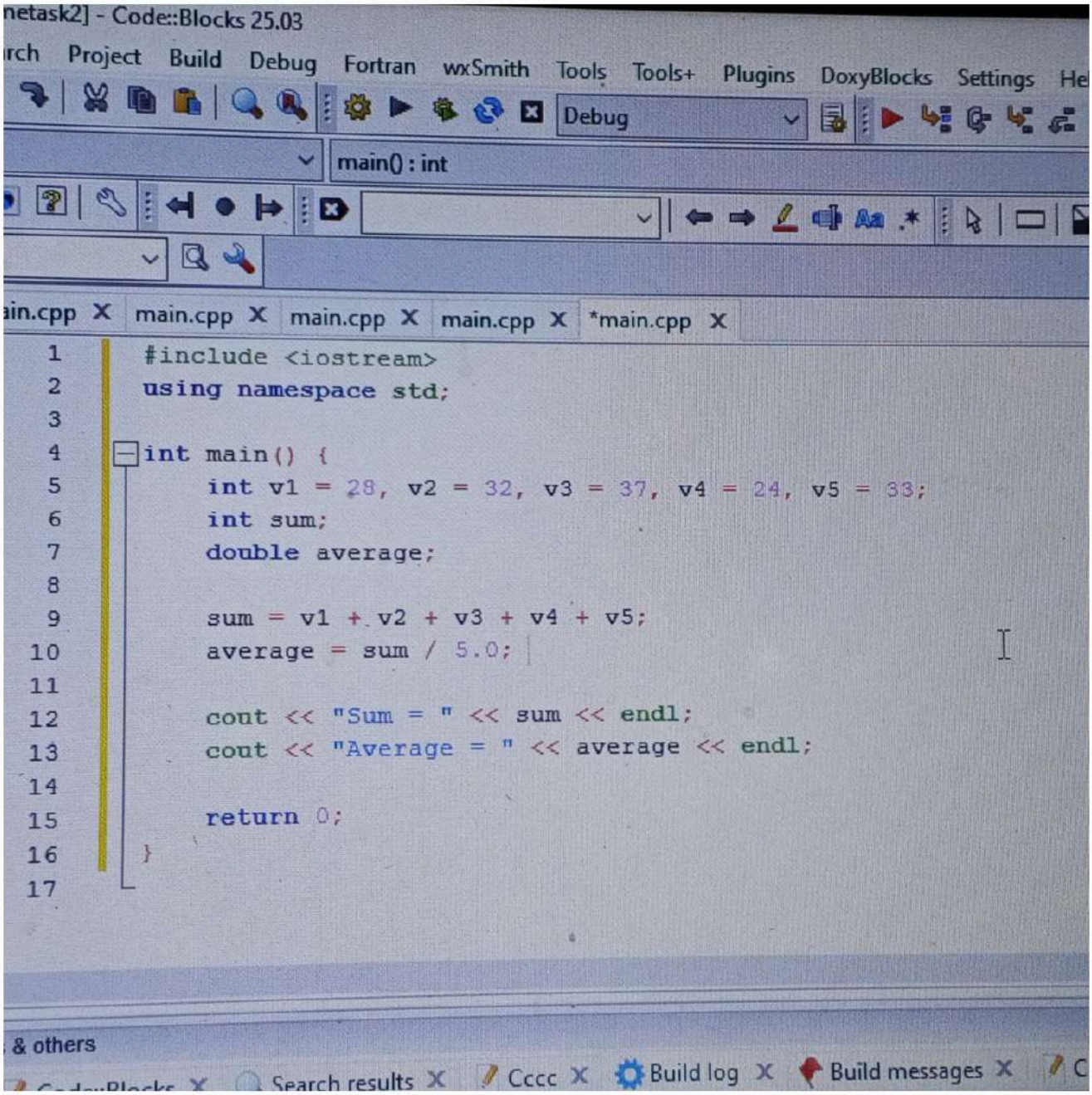
Lab-03 Home task 2

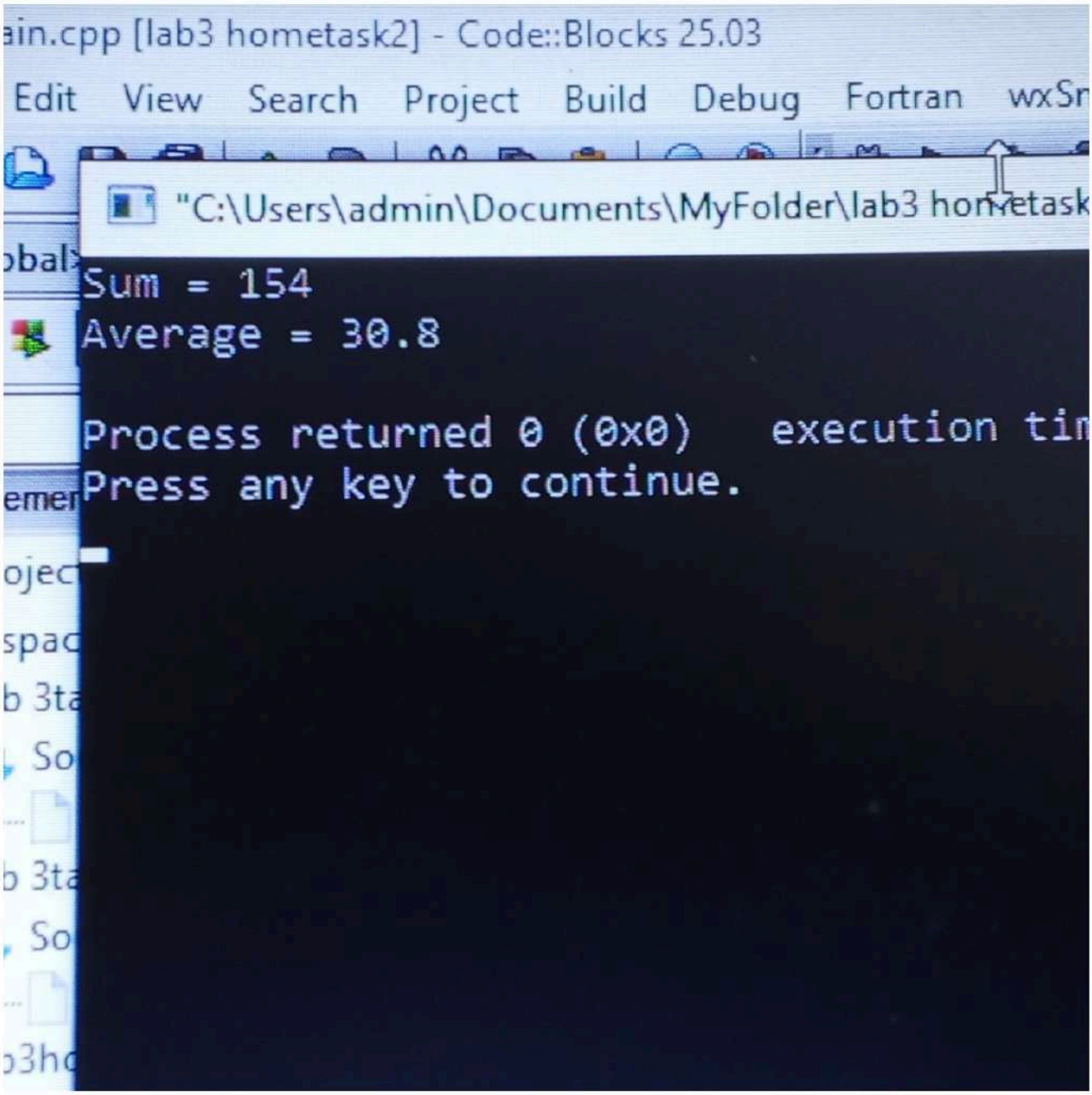
Date:

```
#include<iostream>
using namespace std;
int main () {
    int v1=28, v2=32, v3=37, v4=24, v5=33;
    int sum ;
    double average;
    sum = v1 + v2 + v3 + v4 + v5;
    average = sum / 5.0;
    cout << "sum= " << sum <<endl;
    cout << "Average = " << average <<endl;
    return 0;
}
```

output

sum=154
Average = 30.8


netask2] - Code::Blocks 25.03
Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings
Debug
main() : int
main.cpp
main.cpp
main.cpp
*main.cpp
#include <iostream>
using namespace std;
int main() {
int v1 = 28, v2 = 32, v3 = 37, v4 = 24, v5 = 33;
int sum;
double average;
sum = v1 + v2 + v3 + v4 + v5;
average = sum / 5.0;
cout << "Sum = " << sum << endl;
cout << "Average = " << average << endl;
return 0;
}
& others
Search results
Cccc
Build log
Build messages


ain.cpp [lab3 hometask2] - Code::Blocks 25.03
Edit View Search Project Build Debug Fortran wxSm
"C:\Users\admin\Documents\MyFolder\lab3 hometask
Sum = 154
Average = 30.8
Process returned 0 (0x0) execution tim
Press any key to continue.